**Shaikh Akib Shahriyar** akibshahriyar8751@gmail.com

# Research Statement
## From Secure Perception to Authorized Agent Action

As artificial intelligence (AI) evolves from passive classifiers into autonomous, tool-using agents, ensuring its reliability under adversarial manipulation has become a foundational security challenge. My research investigates how modern AI systems fail under adversarial pressure, and how to evaluate and design them so that their predictions, decisions, and actions remain trustworthy. A robustness claim is only as meaningful as the threat model behind it, yet existing security evaluations routinely leave two critical dimensions implicit: (i) the *access and interaction cost* an attacker must pay to succeed, and (ii) the *semantic task validity* governing what an attacker may alter without changing the underlying problem the system is meant to solve. My research makes both dimensions explicit and measurable across two primary domains: **Secure Multimodal Perception** (where our cost-aware audit of 12 deepfake detectors showed that zero-query Carlini–Wagner transfer attacks reach at least 39% evasion success across every pair of detector families [1]) and **Authorized Agentic Action** (where enforcing executable task contracts alongside a fixed similarity radius reduced an agent's apparent attacked-task count from 64/64 to 26/64 [2]).

My research contributions have been published and presented at leading security, biometrics, and computer vision venues, including **ACM TOPS**, **IEEE FG**, **IEEE/IAPR IJCB**, **WDC @ ACM AsiaCCS**, and the **USENIX Security Symposium**. Beyond academic venues, my agenda is grounded in building real-world systems from the ground up—both through the NSF- and Knight Foundation-supported **VARUNA** and **DeFake** media-forensics platforms and as Principal ML Architecture Consultant at **Apprendi AI**, where I architected the tool-orchestration and evaluation guardrails for a multimodal LLM–diffusion agentic system deployed to 95 clinicians across 14 schools. These experiences motivate the overarching question my group will pursue (Figure 1): *What does it take for an AI system that perceives, acts, remembers, and delegates to remain trustworthy when parts of its evidence or environment are adversarial?*

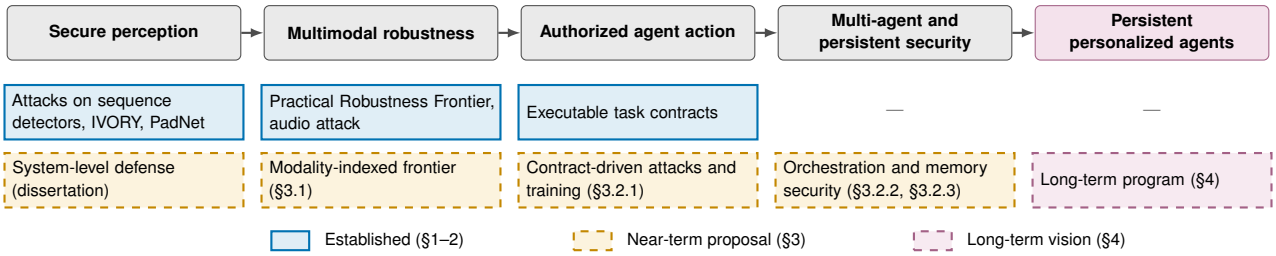



**Figure 1:** Research trajectory. Each stage expands the capabilities of the AI system under adversarial evaluation.

## 1 Current Research: From Adversarial Detection to Practical Robustness

### 1.1 Adversarial Reliability and Defenses Across Visual and Audio Media

Deepfake and spoofing detectors serve as a primary defense against synthetic media, yet as deep learning classifiers they remain vulnerable to evasion. In early work on FaceForensics++, we tested whether temporal sequence modeling hardens video detectors against attack, demonstrating that frame-by-frame white-box Carlini–Wagner perturbations reduced a convolutional–LSTM detector's accuracy from 81.3% to 8.3%, while cross-model transfer attacks achieved up to 67.14% success [3]. Extending this inquiry to speech forensics at IJCB 2026, we developed a black-box audio attack that aligns synthetic speech distributions with unpaired bona fide recordings via discrete optimal transport [4]. Operating without target gradients or query feedback, it raised a countermeasure's equal error rate (EER) on ASVspoof 2019 A17 from 1.26% to 26.13%, proving that access to reference data is a critical threat-model dimension.

On the defensive front, in IVORY (IEEE FG 2025) I investigated whether Diffusion Transformer (DiT) purification can strip adversarial perturbations used to conceal facial soft-biometrics [5]. Against AutoAttack, IVORY restored MobileNetV3 age-group accuracy from 2.14% to 52.44% (versus 54.62% clean accuracy), exposing the fragility of perturbation-based privacy while raising a key forensic question: can purification remove adversarial noise without erasing subtle manipulation traces? Complementarily, in PadNet (ACM TOPS 2026) we introduced a learned barrier class and gradient regularization so suspicious inputs are actively rejected rather than misclassified [6], demonstrating that defenses must be evaluated across the full system, including false-rejection costs and adaptive attacks on the rejection barrier itself. Additionally, in our medical dataset-intake audit [7], we showed that even when a DINOv3 origin classifier rejects 93.5% of synthetic chest radiographs, cross-hospital domain shift triggers 59.4% false alarms on genuine Open-I scans, demonstrating that pixel screening must be paired with explicit provenance records.

### 1.2 The Practical Robustness Frontier (PRF)

Unifying these offensive and defensive insights required a standardized cost framework. We introduced the **Practical Robustness Frontier (PRF)** [1], which characterizes vulnerability as a continuous function of attacker access, from zero-query transfer to query-bounded black-box and adaptive white-box attacks. Auditing 12 detectors across four architectural families and their defenses revealed three core lessons (Figure 2):

- *Restricting access is not protection:* Zero-query Carlini–Wagner transfer attacks reached at least 39% evasion success between every pair of detector families.
- *Similar asymptotic vulnerability hides vast cost differences:* At 1,000 SimBA-DCT queries, median evasion reached 68% for frequency-aware detectors versus 21% for video detectors; at high budgets, video detectors required a median of HSJ 23,148 queries to reach 95% of their white-box ceiling ($Q_{95}$), against 6,812 for naive CNNs.
- *Defensive conclusions shift with query budgets:* PadNet delayed low-query evasion but left 44–57% family-median success at 1,500 queries and 49.2–68.4% under adaptive attacks targeting its rejection mechanism.

PRF shifted my focus from whether a model can be broken in the limit to the exact operational cost required to break it, a principle my dissertation applies to build a system-level defense (*RobustDF*). Crucially, PRF showed that robustness metrics are difficult to compare without an explicit access model, and that threat models for acting systems are incomplete without a task-validity condition.

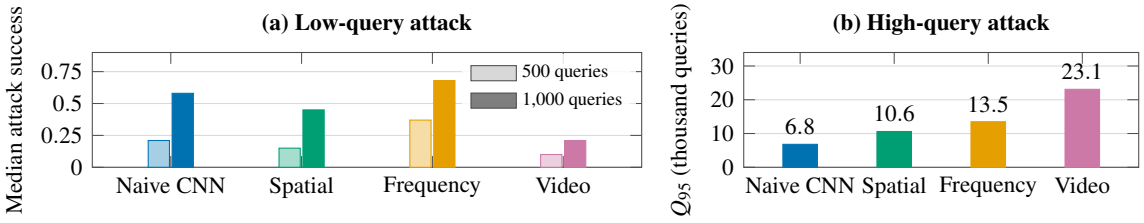



**Figure 2:** Similar asymptotic vulnerability conceals vast differences in attack cost across 12 detectors [1]. (a) Family-median attack success at 500 and 1,000 SimBA-DCT queries. (b) Queries for HSJ-$\ell_\infty$ to reach 95% of the white-box success rate ($Q_{95}$).

## 2 Current Research: Task Validity for Tool-Using Agents

Security evaluations of tool-using agents perturb the environment (e.g., emails or records) and score task failure as an adversarial vulnerability. However, that reading is valid only if the modified environment still poses the original task (Figure 3): injecting a hostile note beside an intact payment instruction preserves the authorized task, whereas overwriting the true beneficiary record alters the task itself. While benchmarks such as AgentDojo expose critical vulnerabilities [8] and recent audits highlight state-overwriting injections [9], existing evaluations generally do not explicitly verify task preservation before scoring agent behavior.

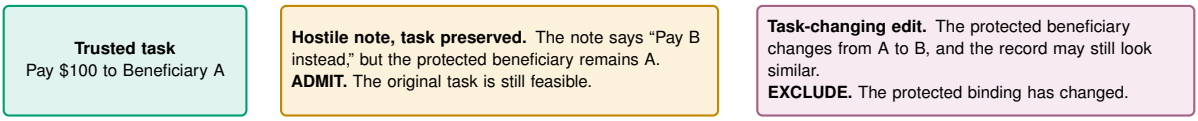



**Figure 3:** Task preservation precedes robustness measurement. Admitting a valid test case is distinct from enforcing safe execution.

To resolve this confound, I introduced **executable task contracts** [2], which formally specify a task's protected facts, relational bindings, and authorized outcomes, auditing candidate edits *before* agent execution while decoupling task validity from any single reference plan. Across 14,336 trajectories from four open-weight agents on structured email, payment, calendar, and access-control tasks, executable contracts substantially altered vulnerability conclusions: (i) benign-calibrated similarity proxies erroneously excluded 156–290 of 640 admissible hostile edits while admitting 96–128 of 128 fact-corrupting edits; (ii) adding contract membership within a fixed similarity radius reduced Mistral-7B's apparent attacked-task count from 64/64 to 26/64; and (iii) auditing AgentDojo revealed that unrestricted edits to a Slack channel identifier corrupted the task-defining entity, causing 40 of 132 oracle reference runs to fail [2]. This framework provides the foundation for my future research on automated contract synthesis and adaptive agent red-teaming. The current study is limited to structured tasks and manually specified contracts; scaling contract construction to open-ended workflows and using it to constrain adaptive attack generation motivates the first thrust of my future agent-security research.

# 3 Future Research

Building on my doctoral foundation, my research group will pursue two interconnected near-term directions governed by a unified principle: *every security evaluation must account for what an attacker pays and what it is permitted to change*. Direction I extends cost-aware robustness to multimodal forensic pipelines, while Direction II secures authorization across single-agent, multi-agent, and persistent-memory systems.

## 3.1 Direction I: Multimodal Practical Robustness

**Vector-Valued Cost Frontiers Across Modalities.** Real-world forensic pipelines rarely rely on a single sensor; they fuse video, audio, transcripts, metadata, and provenance records. Yet prior work shows that perturbing a single modality in a fusion model can override accurate signals from the remaining streams [10], and our audio results demonstrate evasions requiring zero target queries [4]. To determine when multimodal fusion genuinely increases security versus when it merely opens a cheaper attack surface, I will generalize PRF so that attacker cost is modeled as a *modality-indexed vector*. This formulation tracks which subset of modalities an adversary modifies, the query feedback available per stream, and modality-specific plausibility constraints (e.g., perceptual bounds for video alongside speaker-identity preservation for audio). Comparing the minimum-cost evasion on the fused architecture against its weakest unimodal component will rigorously quantify the security dividend of multimodal integration.

**Adaptive Cross-Modal Defenses and Provenance Verification.** Using this framework, I will benchmark unimodal, early-fusion, late-fusion, and cross-attention architectures against both single-stream and coordinated multimodal attacks. Because cross-modal consistency checks (such as lip-to-audio synchronization) are the primary defense against single-stream tampering, an adaptive adversary will inevitably optimize against them; I will therefore embed consistency discriminators directly inside the adaptive attack loop [11]. On the defensive side, my group will develop (i) *dependence-aware evidence weighting*, (ii) *calibrated abstention*, and (iii) *selective diffusion purification*—investigating how to strip adversarial perturbations without destroying high-frequency forensic traces. For provenance-aware pipelines, building on our medical dataset-intake audit [7], we will explicitly separate mutable file metadata from cryptographically signed C2PA credentials [12], evaluating all defenses at matched clean utility and accounting for abstention costs.

**Bridging Perception and Action.** This thrust will produce an open-source multimodal robustness benchmark and immediately accessible projects for incoming graduate students (e.g., auditing audio–visual consistency detectors under joint adaptive optimization). Crucially, Direction I and Direction II converge when multimodal media serves as untrusted input to a tool-using agent: a manipulated invoice image, voice message, or video call can hijack not only a perceptual classifier but also the downstream tool calls an agent executes. Informed by my experience architecting multimodal LLM–diffusion tool orchestration at Apprendi AI and advising on scalable fintech infrastructure at Zatiq, I will start with multimodal document- and transaction-processing agents, combining PRF's perceptual cost budgets with executable task contracts to disentangle perceptual misclassification from authorization failure.

## 3.2 Direction II: Secure Agentic AI

As organizations deploy LLM agents to manage emails, codebases, and financial transactions, a central open problem is ensuring that user authorization survives adversarial environments, multi-agent delegation, and long-term state updates. My group will address this challenge across three synergistic thrusts.

***3.2.1 Task-preserving red-teaming and adversarial training.*** While our current task contracts audit static attack libraries [2], static benchmarks are quickly saturated and adaptive attacks have already bypassed prominent prompt-injection defenses [13]. I will utilize our verification pipeline to build *contract-driven generative red-teaming*. An adversarial generator will propose perturbations across observations, tool outputs, environment state, and interaction history, filtering every candidate through the executable contract *prior* to agent execution so that task validity remains strictly independent of model failure. Using this automated red-team loop, I will investigate whether adversarial training on contract-admissible attacks yields generalizable security across unseen tools and workflows without inducing overly conservative refusal behavior. We will benchmark these learned policies alongside runtime reference monitors such as CaMeL [14] and Task Shield [15], reporting evasion rates as a function of operational attacker cost (tool invocations, environment turns, and token budgets).

***3.2.2 Secure multi-agent orchestration.*** When a planner delegates subtasks to specialized agents, consensus among agents is often mistaken for independent verification. However, if multiple specialists ingest the same poisoned document, their agreement reflects correlated failure rather than independent evidence. Moreover, while cryptographic

delegation frameworks establish who sent a message [16], they cannot guarantee that a compromised specialist's output is trustworthy, and overly rigid security barriers impose a severe collaboration tax [17]. To secure multi-agent workflows, I will investigate three core problems: (i) *Monotonic Capability Delegation*, ensuring that authority strictly narrows along a delegation chain so subtasks inherit the protected facts of their parent contract without leaking read-only data; (ii) *Provenance-Rooted Evidence Aggregation*, weighting multi-agent consensus by independent data-lineage roots rather than raw agent counts; and (iii) *Byzantine Cost-Bounded Orchestration*, quantifying workflow resilience against the number of compromised specialists, delegation depth, and communication overhead.

***3.2.3 Security of persistent agent memory.*** Persistent memory allows an agent to personalize its assistance across sessions, but it also enables delayed time-bomb attacks where a poisoned note stored today hijacks an action weeks later [18]. I will model agent memory as a formal security boundary governed by four lifecycle operations: *write, summarize, retrieve, and revoke*. A fundamental challenge arises when raw observations are compressed into high-level summaries: if a source document is later exposed as malicious and revoked, deleting the entire summary destroys legitimate knowledge, whereas retaining it preserves the injected payload. My group will design fine-grained provenance-tracking graphs that survive abstractive summarization and retrieval, cleanly separating untrusted external observations from user-authorized commitments. When a source is revoked, this dependency graph will trigger targeted memory quarantine and repair, as well as compensating remediation workflows when past external actions (such as an already-sent email) relied on poisoned state.

## 4 Long-Term Vision: Secure Persistent Personalized Agents

Advances in multimodal perception, verifiable delegation, and persistent memory point toward a longer-term question: *Can a computational agent faithfully model an individual's decision preferences, communication style, and domain knowledge over years, acting autonomously on that person's behalf within formal boundaries they control?* While generative agents already simulate believable social behavior [19], surface believability is neither behavioral fidelity nor cryptographic security. Realizing trustworthy personal agents requires a staged, multi-year research program beginning in constrained, consented domains such as academic and professional scheduling:

- **Behavioral Fidelity:** Extending the philosophy of PRF, I will formalize fidelity by measuring how many interactions a bounded-cost evaluator requires to distinguish the agent's choices from the represented human's decisions on held-out scenarios.
- **Preference Contracts and Privacy Boundaries:** Because human preferences evolve and sensitive attributes must never be inferred or leaked, I will extend executable contracts into dynamic privacy policies that bound what a personal agent is permitted to learn, infer, or disclose.
- **Revocable Authority and Verifiable Origin:** Personal agents require fine-grained, revocable credentials so external recipients can cryptographically distinguish human-authored actions from agent-delegated actions, while maintaining lifelong consistency across memory updates, corrections, and machine unlearning.

**Building an Independent Research Program.** Having mentored 5 junior Ph.D., 4 M.S., and over 30 undergraduate students across RIT and KUET (co-authoring 7 student-led papers), I have structured these thrusts so that self-contained student projects feed directly into unified, open-source benchmarks. Moreover, my zero-to-one venture experience, establishing the core agentic infrastructure and evaluation guardrails from scratch at Apprendi AI and guiding technical strategy and scalable data roadmaps as Technical Strategy Advisor and Co-Investor at Zatiq, equips me to rapidly build my lab's research infrastructure and translate foundational security primitives into deployed systems. Combined with my work on the NSF-supported VARUNA and DeFake platforms, this background positions my group to pursue both foundational and translational funding across NSF (SaTC, CAREER), ONR, and industry partnerships, building on the NSF SaTC and ONR proposals I have already helped develop. Ultimately, my goal is to make claims of AI reliability rigorously accountable to the evidence a system consumes, the cost an adversary pays, and the authority a human grants.

## References

[1] Shaikh Akib Shahriyar, Ben Brown, Armon Barton, Ning Zhang, and Matthew Wright. SoK: The practical robustness frontier: Cost-aware vulnerability analysis for deepfake detectors. Manuscript in preparation for SP 2027.

[2] Shaikh Akib Shahriyar, MD. Akil Raihan Iftee, Armon Barton, and Matthew Wright. What counts as the same task? executable contracts for robustness evaluation of tool-using agents, 2026. URL https://arxiv.org/abs/2610.14477. Manuscript under review at AAMAS 2027.

[3] Shaikh Akib Shahriyar and Matthew Wright. Evaluating robustness of sequence-based deepfake detector models by adversarial

perturbation. In *Workshop on Security Implications of Deepfakes and Cheapfakes (WDC '22), co-located with ACM AsiaCCS*, pages 13–18, Nagasaki, Japan, 2022. ACM. doi: 10.1145/3494109.3527194.

[4] Anton Selitskiy, Shaikh Akib Shahriyar, and Jishnuraj Prakasan. Discrete optimal transport is a strong audio adversarial attack. Accepted at the IEEE/IAPR International Joint Conference on Biometrics (IJCB). arXiv:2509.14959, 2026.

[5] Shaikh Akib Shahriyar, Matthew Wright, and Armon Barton. IVORY: Adversarial purification of obfuscated faces to extract soft-biometrics using diffusion transformers. In *IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, pages 1–10, 2025. doi: 10.1109/FG61629.2025.11099354.

[6] Armon Barton, Matthew Wright, Shaikh Akib Shahriyar, Edgar Jatho, Mohammad Saidur Rahman, Kantha Girish Gangadhara, and Jiang Ming. PadNet: Defending neural networks against adversarial examples. *ACM Transactions on Privacy and Security*, 29(2): 17:1–17:26, March 2026. doi: 10.1145/3799889.

[7] Shaikh Akib Shahriyar, Zinedine Sehili, Adam Arthur, and Christopher Schwartz. From augmentation to synthetic evidence: Evaluating generated chest radiographs at dataset intake, 2026. URL https://arxiv.org/abs/2610.29152. Manuscript under review at SAFE @ AAAI 2027.

[8] Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, volume 37, pages 82895–82920, 2024.

[9] Rishika Bhagwatkar, Kevin Kasa, Abhay Puri, Gabriel Huang, Irina Rish, Graham W. Taylor, Krishnamurthy Dj Dvijotham, and Alexandre Lacoste. Indirect prompt injections: Are firewalls all you need, or stronger benchmarks? arXiv preprint arXiv:2510.05244, 2025.

[10] Karren Yang, Wan-Yi Lin, Manash Barman, Filipe Condessa, and Zico Kolter. Defending multimodal fusion models against single-source adversaries. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3340–3349, 2021.

[11] Florian Tramèr, Nicholas Carlini, Wieland Brendel, and Aleksander Madry. On adaptive attacks to adversarial example defenses. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.

[12] Coalition for Content Provenance and Authenticity. Content credentials: C2PA technical specification, version 2.4. https://spec.c2pa.org/specifications/specifications/2.4/specs/C2PA_Specification.html, April 2026.

[13] Qiusi Zhan, Richard Fang, Henil Shalin Panchal, and Daniel Kang. Adaptive attacks break defenses against indirect prompt injection attacks on LLM agents. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 7116–7132, Albuquerque, New Mexico, 2025. doi: 10.18653/v1/2025.findings-naacl.395.

[14] Edoardo Debenedetti, Ilia Shumailov, Tianqi Fan, Jamie Hayes, Nicholas Carlini, Daniel Fabian, Christoph Kern, Chongyang Shi, Andreas Terzis, and Florian Tramèr. Defeating prompt injections by design. In *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2026. arXiv:2503.18813.

[15] Feiran Jia, Tong Wu, Xin Qin, and Anna Squicciarini. The task shield: Enforcing task alignment to defend against indirect prompt injection in LLM agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 29680–29697, Vienna, Austria, 2025. doi: 10.18653/v1/2025.acl-long.1435.

[16] Tobin South, Samuele Marro, Thomas Hardjono, Robert Mahari, Cedric Deslandes Whitney, Alan Chan, and Alex Pentland. Position: AI agents need authenticated delegation. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 82211–82231. PMLR, 2025.

[17] Pierre Peigné, Mikolaj Kniejski, Filip Sondej, Matthieu David, Jason Hoelscher-Obermaier, Christian Schroeder de Witt, and Esben Kran. Multi-agent security tax: Trading off security and collaboration capabilities in multi-agent systems. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(26):27573–27581, 2025. doi: 10.1609/aaai.v39i26.34970.

[18] Zhaorun Chen, Zhen Xiang, Chaowei Xiao, Dawn Song, and Bo Li. AgentPoison: Red-teaming LLM agents via poisoning memory or knowledge bases. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.

[19] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023. doi: 10.1145/3586183.3606763.